

Central branching is the only maximality obstruction for zero-diagonal MASAs in a type II_1 factor

Automated mathematical discovery packet

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Status. Candidate partial result, likely valid, subject to human review. The full MASA question remains open.

1 Source problem

The source is K. Dykema and P. Skoufranis, *Numerical Ranges in II_1 Factors*, arXiv:1503.05766 [1]. On physical PDF page 19, immediately before Lemma 6.1, the authors write:

“Unfortunately, we do not obtain true ‘diagonals’ as we do not know if one can guarantee $\mathcal{A}$ in the following technical lemma (whose proof is a generalization of a matricial result) is a MASA.”

The lemma then says that if $(\mathcal{M}, \tau)$ is a type II_1 factor and $\tau(T) = 0$, there is a diffuse abelian von Neumann subalgebra $\mathcal{A} \subset \mathcal{M}$ such that $E_{\mathcal{A}}(T) = 0$. Thus the question is:

Can $\mathcal{A}$ always be chosen maximal abelian?

6. NUMERICAL RANGES AND DIAGONALS

In this our final section, we desire description of when a scalar belongs to the α -numerical range of an operator based on the possible ‘diagonals’ of an operator. Our characterization is similar to that for k -numerical ranges of matrices found in [14 Theorem 2.4]. Unfortunately, we do not obtain true ‘diagonals’ as we do not know if one can guarantee $\mathcal{A}$ in the following technical lemma (whose proof is a generalization of a matricial result) is a MASA.

Lemma 6.1. *Let $(\mathfrak{M}, \tau)$ be a type II_1 factor and let $T \in \mathfrak{M}$ be such that $\tau(T) = 0$. Then there exists a diffuse abelian von Neumann subalgebra $\mathcal{A}$ of $\mathfrak{M}$ such that $E_{\mathcal{A}}(T) = 0$, where $E_{\mathcal{A}} : \mathfrak{M} \rightarrow \mathcal{A}$ is the normal conditional expectation.*

Proof. Notice $0 \in V_1(T) \subseteq V_{\frac{1}{2}}(T)$. Hence there exists a projection $P \in \mathfrak{M}$ such that $\tau(P) = \frac{1}{2}$ and $\tau(TP) = 0$. Similarly, $\tau(T(I_{\mathfrak{M}} - P)) = 0$. By repeating this argument in $P\mathfrak{M}P$ and $(I_{\mathfrak{M}} - P)\mathfrak{M}(I_{\mathfrak{M}} - P)$, we obtain four projections $\{P_k\}_{k=1}^4$ such that P_k commutes with P and $I_{\mathfrak{M}} - P$, $\tau(P_k) = \frac{1}{4}$, and $\tau(TP_k) = 0$ for all k . By continuing to repeat the first argument on each compression and by taking the von Neumann algebra generated by these projections, the desired diffuse abelian von Neumann subalgebra of $\mathfrak{M}$ is obtained. ■

Figure 1: Exact source passage and Lemma 6.1, rendered from the stored arXiv source (physical PDF page 19).

2 Definitions

All conditional expectations are the unique trace-preserving normal expectations. Call an abelian von Neumann algebra $\mathcal{A} \subset \mathcal{M}$ *zero-diagonal for T* if $E_{\mathcal{A}}(T) = 0$. This is equivalent to

$$\tau(Ta) = 0 \quad (a \in \mathcal{A}).$$

For such $\mathcal{A}$, set

$$\mathcal{N} = \mathcal{A}' \cap \mathcal{M}, \quad \mathcal{Z} = Z(\mathcal{N}), \quad g = E_{\mathcal{Z}}(T).$$

Since every element of $\mathcal{N}$ commutes with $\mathcal{A}$, one has $\mathcal{A} \subset \mathcal{Z}$.

We use the following concrete form of relative atomlessness. The inclusion $\mathcal{A} \subset \mathcal{Z}$ is *relatively atomless* if, after identifying the two abelian algebras with nested sub- σ -algebras of a standard probability space, the larger measure algebra has no atom relative to the smaller one. The inclusion $\mathcal{A} \subset \mathcal{Z}$ has the *conditional halving property for g* if there is a projection $q \in \mathcal{Z}$ such that

$$E_{\mathcal{A}}(q) = \frac{1}{2}1, \quad E_{\mathcal{A}}(gq) = 0.$$

The conditional Lyapunov theorem [4] implies this property whenever the abelian extension $\mathcal{A} \subset \mathcal{Z}$ is relatively atomless.

3 Proof intuition

First maximize, by inclusion, an abelian algebra on which T has zero diagonal. Such a maximal algebra cannot have an atom: the source lemma can be inserted into the corresponding II_1 corner and gives a strict zero-diagonal enlargement.

The only place where maximality might still fall short of being a MASA is the relative commutant $\mathcal{N} = \mathcal{A}' \cap \mathcal{M}$. If the central part of T in $\mathcal{N}$ vanishes, then every factor fiber of $\mathcal{N}$ contains a zero-diagonal abelian algebra. A measurable choice of these fiber algebras assembles to a zero-diagonal extension containing the center, contradicting maximality. Hence a non-MASA maximal algebra must carry a nonzero central remnant g . Even that remnant can be split away when the central extension is relatively atomless. The sole remaining configuration is therefore nonzero, conditionally atomic central branching.

4 The center-valued zero-diagonal lemma

Lemma 1 (center-valued zero diagonal). *Let $(\mathcal{N}, \tau)$ be a finite von Neumann algebra with separable predual, let $\mathcal{Z} = Z(\mathcal{N})$, and let $S \in \mathcal{N}$ satisfy $E_{\mathcal{Z}}(S) = 0$. Then there is an abelian von Neumann algebra $\mathcal{D}$ such that*

$$\mathcal{Z} \subset \mathcal{D} \subset \mathcal{N}, \quad E_{\mathcal{D}}(S) = 0.$$

If $\mathcal{N}$ is nonabelian on a nonzero central summand, $\mathcal{D}$ can be chosen to strictly contain $\mathcal{Z}$ on that summand.

Proof. Decompose $\mathcal{N}$ over its center as a standard measurable field of finite factors,

$$\mathcal{N} = \int_X^{\oplus} \mathcal{N}_x d\mu(x), \quad S = \int_X^{\oplus} S_x d\mu(x).$$

The hypothesis says that the normalized scalar trace $\tau_x(S_x)$ is zero for almost every x .

If $\mathcal{N}_x = M_n(\mathbb{C})$, choose a unit vector e_1 with $\langle S_x e_1, e_1 \rangle = 0$. Such a vector exists because the numerical range is convex, contains the spectrum, and therefore contains the average of the eigenvalues, namely zero. Compressing to $e_1^\perp$ leaves trace zero; induction gives an orthonormal basis in which every diagonal entry is zero. The corresponding diagonal algebra $\mathcal{D}_x$ satisfies $E_{\mathcal{D}_x}(S_x) = 0$. For a type II_1 fiber, Lemma 6.1 of [1] supplies a diffuse abelian $\mathcal{D}_x$ with the same property. A scalar fiber has $S_x = 0$, and we put $\mathcal{D}_x = \mathbb{C}$ there.

It remains to make the choices measurable. On the unit balls of a separable measurable field, the projection relation, normalized trace, compression, and the equations $\tau_x(S_x p) = 0$ are Borel conditions and are closed in the 2-norm on bounded sets. The nonempty solution fields therefore admit measurable selectors by the Kuratowski–Ryll–Nardzewski theorem [2]. Applying this recursively to the finite-dimensional basis choices and to the dyadic projection tree in the proof of Lemma 6.1 produces a measurable field $x \mapsto \mathcal{D}_x$. Direct-integral theory [3] then gives

$$\mathcal{D} = \int_X^\oplus \mathcal{D}_x d\mu(x).$$

It contains the diagonal center $L^\infty(X)$, and the fiberwise identities give $E_{\mathcal{D}}(S) = 0$. On every nonabelian factor fiber the construction contains a nonscalar projection, so measurable selection on any positive-measure nonabelian stratum makes $\mathcal{D}$ strictly larger than $\mathcal{Z}$ there. $\square$

5 Main partial theorem

Theorem 1 (central-obstruction reduction). *Let $(\mathcal{M}, \tau)$ be a type II_1 factor with separable predual and let $T \in \mathcal{M}$ satisfy $\tau(T) = 0$.*

- (i) *There is an abelian zero-diagonal algebra $\mathcal{A}$ maximal by inclusion, and every such maximal $\mathcal{A}$ is diffuse.*
- (ii) *For a maximal $\mathcal{A}$, put $\mathcal{N} = \mathcal{A}' \cap \mathcal{M}$ and $\mathcal{Z} = Z(\mathcal{N})$. If $E_{\mathcal{Z}}(T) = 0$, then $\mathcal{A}$ is a MASA.*
- (iii) *In particular, $Z(\mathcal{A}' \cap \mathcal{M}) = \mathcal{A}$ implies that $\mathcal{A}$ is a MASA.*
- (iv) *If $\mathcal{A}$ is not a MASA, then $\mathcal{A} \subset \mathcal{Z}$ fails the conditional halving property for $g = E_{\mathcal{Z}}(T)$. Consequently, a maximal $\mathcal{A}$ whose strict central extension is relatively atomless must be a MASA.*

Thus, if a maximal zero-diagonal $\mathcal{A}$ is not a MASA, then necessarily

$$\mathcal{A} \subsetneq Z(\mathcal{A}' \cap \mathcal{M}), \quad E_{Z(\mathcal{A}' \cap \mathcal{M})}(T) \neq 0,$$

and the central extension fails relative atomlessness.

Proof. Lemma 6.1 of [1] makes the family of zero-diagonal abelian algebras nonempty. Let $(\mathcal{A}_i)$ be a chain in this family and put $\mathcal{A} = (\bigcup_i \mathcal{A}_i)''$. Every $a \in \mathcal{A}$ is an L^2 limit, uniformly bounded after a standard truncation, of elements from the union. Hence $\tau(Ta) = 0$, so $E_{\mathcal{A}}(T) = 0$. Zorn's lemma gives a maximal $\mathcal{A}$.

Suppose that p is an atom of $\mathcal{A}$. Since $E_{\mathcal{A}}(T) = 0$, $\tau(pTp) = \tau(Tp) = 0$. The corner $p\mathcal{M}p$ is a type II_1 factor, so Lemma 6.1 of [1] gives a diffuse abelian algebra $\mathcal{B}_p \subset p\mathcal{M}p$ with $E_{\mathcal{B}_p}(pTp) = 0$. Then

$$(1 - p)\mathcal{A} \oplus \mathcal{B}_p$$

strictly contains $\mathcal{A}$ and still pairs to zero with T , contradicting maximality. Thus $\mathcal{A}$ is diffuse. This proves (i).

Put $\mathcal{N} = \mathcal{A}' \cap \mathcal{M}$, $\mathcal{Z} = Z(\mathcal{N})$, and $S = E_{\mathcal{N}}(T)$. Conditional-expectation composition gives

$$E_{\mathcal{Z}}(S) = E_{\mathcal{Z}}(T).$$

If this is zero, Lemma 1 gives an abelian $\mathcal{D}$ with $\mathcal{Z} \subset \mathcal{D} \subset \mathcal{N}$ and $E_{\mathcal{D}}(S) = 0$. Since $E_{\mathcal{D}}(T) = E_{\mathcal{D}}(S)$, maximality forces $\mathcal{D} = \mathcal{A}$. But $\mathcal{A} \subset \mathcal{Z} \subset \mathcal{D}$, so $\mathcal{Z} = \mathcal{A}$ as well. If $\mathcal{N}$ were larger than $\mathcal{A}$, either it would be abelian, in which case $\mathcal{N} = \mathcal{Z}$ is already a strict zero-diagonal extension, or it would be nonabelian on a nonzero central summand, where Lemma 1 makes $\mathcal{D}$ strict. Both contradict maximality. Hence $\mathcal{N} = \mathcal{A}$, exactly the MASA condition. This proves (ii), and (iii) follows from $E_{\mathcal{Z}}(T) = E_{\mathcal{A}}(T) = 0$.

Finally suppose the conditional halving property holds. Let $\mathcal{B} = \mathcal{A} \vee \{q\} \subset \mathcal{Z}$. Since $E_{\mathcal{A}}(g) = E_{\mathcal{A}}(T) = 0$, the two central atoms satisfy

$$E_{\mathcal{A}}(gq) = 0, \quad E_{\mathcal{A}}(g(1-q)) = 0.$$

Because $E_{\mathcal{A}}(q) = E_{\mathcal{A}}(1-q) = 1/2$, the explicit two-atom conditional expectation gives $E_{\mathcal{B}}(g) = 0$. Also $E_{\mathcal{B}}(T) = E_{\mathcal{B}}(E_{\mathcal{Z}}(T)) = E_{\mathcal{B}}(g) = 0$. The equality $E_{\mathcal{A}}(q) = 1/2$ forces $q \notin \mathcal{A}$, so $\mathcal{B}$ is a strict zero-diagonal extension. This contradicts maximality and proves the first assertion of (iv). Indeed, for a relatively atomless extension, apply the conditional Lyapunov theorem [4] to the two-coordinate conditional vector measure

$$q \mapsto (E_{\mathcal{A}}(q), E_{\mathcal{A}}(gq)).$$

Its range contains half of the value at 1, namely $(\frac{1}{2}1, \frac{1}{2}E_{\mathcal{A}}(g)) = (\frac{1}{2}1, 0)$. Thus it supplies the required projection q and proves the consequence. $\square$

6 Why the central obstruction cannot simply be deleted

Consider

$$\mathcal{N} = M_2(\mathbb{C}) \oplus \mathbb{C}, \quad \tau = \frac{1}{2} \text{tr}_2 \oplus \frac{1}{2} \text{id}, \quad S = I_2 \oplus (-1).$$

Then $\tau(S) = 0$. A projection has the form $p \oplus e$, where $\text{rank}(p) = r \in \{0, 1, 2\}$ and $e \in \{0, 1\}$. Directly,

$$\tau(S(p \oplus e)) = \frac{r}{4} - \frac{e}{2}.$$

This vanishes only for $(r, e) = (0, 0)$ or $(2, 1)$, namely the zero and identity projections. Thus scalar trace zero in a finite algebra does not force a nontrivial zero-diagonal abelian algebra when the center-valued trace is nonzero. This is an obstruction model, not a counterexample to the factor question.

7 Verification and eight upgrade attempts

The proof was checked step by step in `VERIFIER_REPORT.md`. The finite obstruction script exhausts all six projection-rank cases and prints `VERDICT: PASS`. The eight focused routes were:

1. Zorn maximalization and atom removal (successful);
2. arbitrary MASA extension (fails because expectation need not persist);
3. localization to the relative commutant (successful reduction);

4. measurable factor-fiber construction under $E_{\mathcal{Z}}(T) = 0$ (successful);
5. forcing $Z(\mathcal{A}' \cap \mathcal{M}) = \mathcal{A}$ (no preserving genericity argument found);
6. conditional Lyapunov splitting (successful refinement);
7. direct elimination of the central term (blocked by the finite model);
8. commutator/counterexample routes and a bounded literature audit (no full resolution found).

The detailed attempt log is stored in the run’s `attempts/` directory.

8 Literature and novelty check

The bounded search covered the exact title and arXiv id, the exact phrase “we do not know if one can guarantee”, the author pair, and combinations of “trace zero”, “zero diagonal”, “MASA”, “conditional expectation”, and “type II_1 factor” in arXiv and primary publication pages. Searches also covered later Schur–Horn and multivariable Schur–Horn papers. No paper explicitly resolving the source question was found. Marcoux’s 2022 survey still states the broader C-star-algebra version as Problem 3.3 [5].

Novelty confidence is moderate. The ingredients of Lemma 1 are standard, but this precise maximality/central branching reduction was not found in the bounded search.

9 Limitations and human-review recommendation

The theorem is partial and assumes separable predual in the measurable selection step. It neither proves that the central expectation must vanish nor constructs a factor counterexample when it does not. Human review should focus on the Effros-Borel assembly in Lemma 1 and on the precise conditional Lyapunov hypothesis in part (iv). If those standard measurable steps are accepted, the rest of the proof is elementary.

References

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- [4] E. B. Dynkin and I. V. Evstigneev, *Regular conditional expectations of correspondences*, Theory Probab. Appl. 21 (1977), 325–338; doi:10.1137/1121037.
- [5] L. W. Marcoux, *Five Hilbert Space Problems in Operator Algebras*, Complex Anal. Oper. Theory 16 (2022), article 116.